COHOMOLOGY AND HIGHER DIRECT IMAGES OF A MODULE AGREE WITH THOSE OF THE UNDERLYING ABELIAN SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and covers the bounded below complex case by the same argument; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-modules-abelian also occurs in etale-cohomology.tex. There the statement compares H^i(G, -) for modules over a group ring with H^i(G, -) for modules over the group, which is a different proposition; only the label coincides, so this card counts its own statement.

Allowed standard prerequisites (editorial).

- Compatibility of derived functors along a morphism of ringed spaces. Same chapter, Remark remark-elucidate-lemma, invoked at proof lines 2202-2204. Supplies that one injective resolution computes both derived functors.
- Description of higher direct images by derived pushforward. Same chapter, Lemma lemma-describe-higher-direct-images, invoked at proof lines 2208-2209. Supplies part (2) from part (1).

Source reference. Original lemma 01E4: sources/stacks-cohomology/pinned-original-source.tex, source lines 592–592. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 01F0: sources/stacks-cohomology/pinned-original-source.tex, source lines 2147–2147. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module.*

(1) *The cohomology groups $H^i(U, \mathcal{F})$ for $U \subset X$ open of $\mathcal{F}$ computed as an $\mathcal{O}_X$-module, or computed as an abelian sheaf are identical.*
(2) *Let $f : X \to Y$ be a morphism of ringed spaces. The higher direct images $R^i f_* \mathcal{F}$ of $\mathcal{F}$ computed as an $\mathcal{O}_X$-module, or computed as an abelian sheaf are identical.*

There are similar statements in the case of bounded below complexes of $\mathcal{O}_X$-modules.

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Proof. Consider the morphism of ringed spaces $(X, \mathcal{O}_X) \to (X, \underline{\mathbf{Z}}_X)$ given by the identity on the underlying topological space and by the unique map of sheaves of rings $\underline{\mathbf{Z}}_X \to \mathcal{O}_X$. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Denote $\mathcal{F}_{ab}$ the same sheaf seen as an $\underline{\mathbf{Z}}_X$-module, i.e., seen as a sheaf of abelian groups. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution. By Remark 01F0 we see that $\Gamma(X, \mathcal{I}^\bullet)$ computes both $R\Gamma(X, \mathcal{F})$ and $R\Gamma(X, \mathcal{F}_{ab})$. This proves (1).

To prove (2) we use (1) and Lemma 01E4. The result follows immediately. □